

Student Performance Analysis Report

Data Summary

- Rows: 20
- Columns: 13
- Duplicates: 0

Hypothesis

- n: 20
- pearson_correlation_coefficient: 0.6669176959693142
- pearson_probability: 0.0013192856676435163
- median_hours_split: 2.6
- mean_score_low_hours: 59.8625
- mean_score_high_hours: 78.9
- test_statistics: 3.7432904754115426
- test_probability: 0.0015656909175103418
- interpretation: Correlation(study hours related to scores) Significant (probability value=0.00). Group difference(difference between low and high study hours) Significant (probability value=0.00).

Conclusions

- gender: {'M': 10, 'F': 9}
- region: {'Urban': 11, 'Rural': 8}
- parent_education: {'Graduate': 7, 'HighSchool': 4, 'Postgraduate': 4, 'Illiterate': 3, 'Gone for Milk': 1}

Key Points

- Attendance to Score Relationship: Attendance below 70% average score 55.60, Attendance above 70% average score 75.29.
- More Attendance = More Score, send your child to school
- Correlation(study hours related to scores) Significant (probability value=0.00). Group difference(difference between low and high study hours) Significant (probability value=0.00).

Tables

crosstab_region_passfail

pass	Fail	Pass
region		
Rural	0.125000	0.875000
Urban	0.090909	0.909091

pivot_gender_mean_score

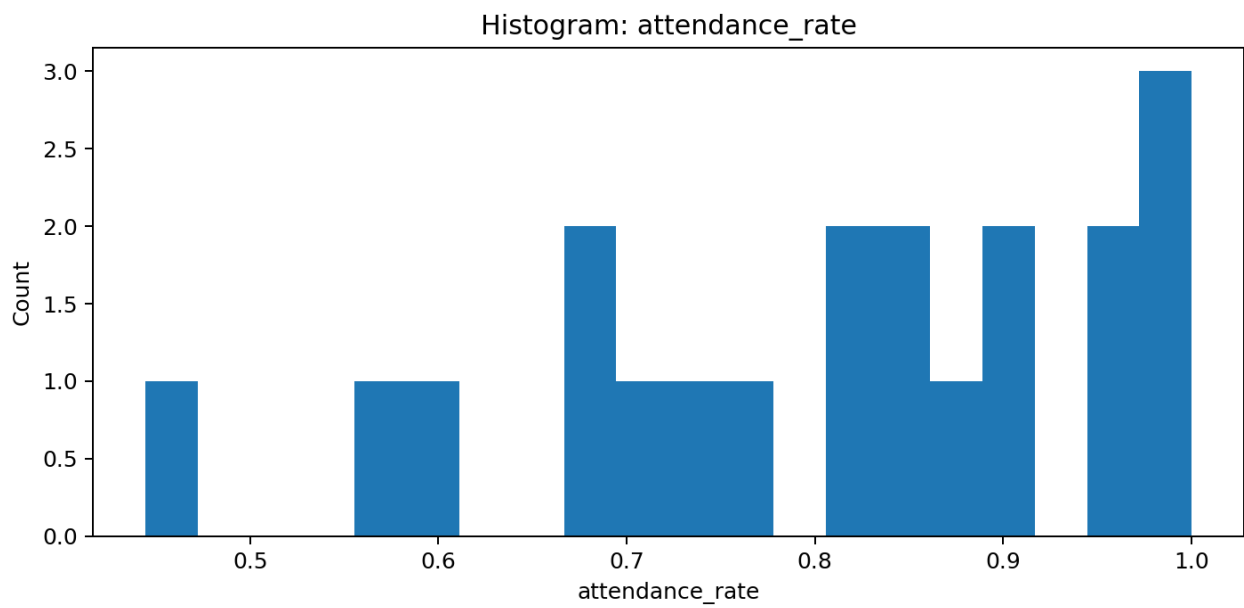
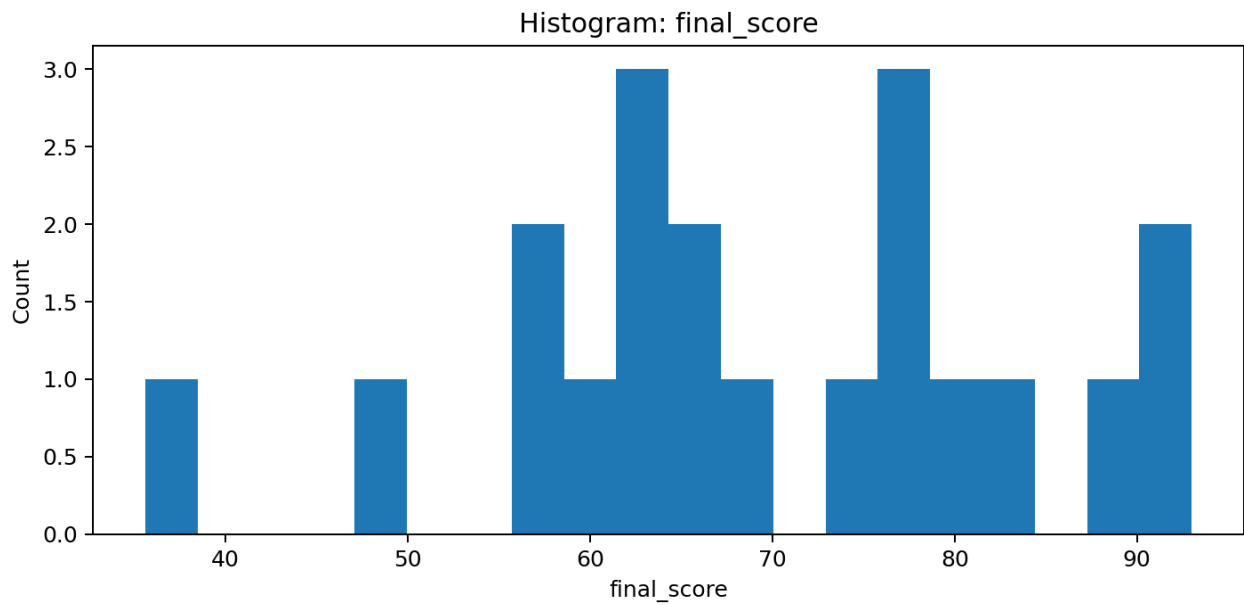
	final_score
gender	
M	68.762500
F	67.666667

pivot_region_mean_score

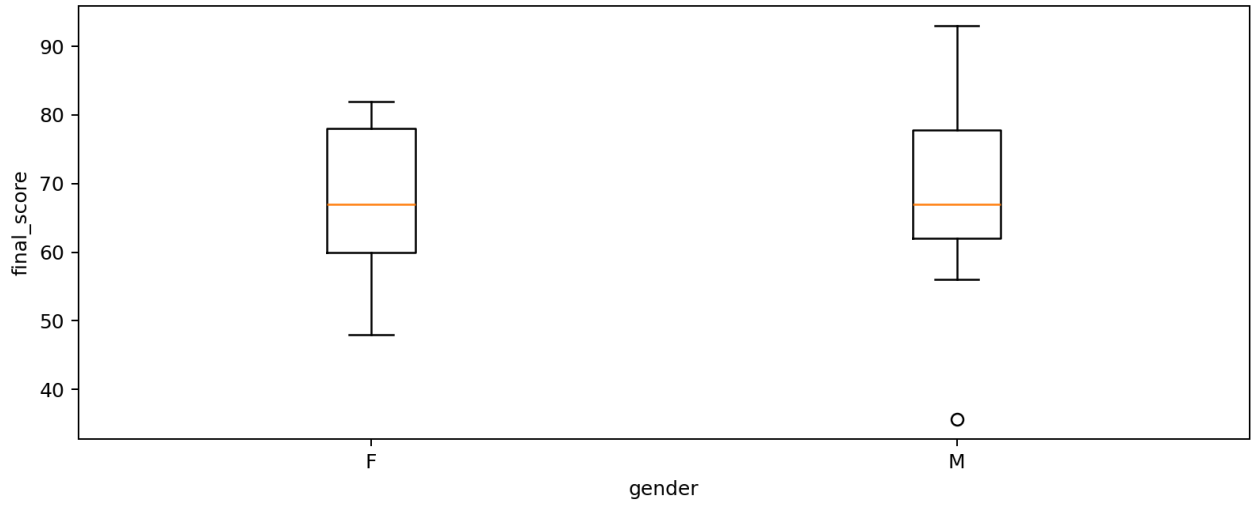
	final_score
region	
Urban	73.545455

	final_score
region	
Rural	60.953125

Visualizations



Boxplot: final_score by gender



Scatter: final_score vs study_hours (trend)

